

Core Student & Teacher Learning Experience Implementation

- **Centralized API Module (src/api/lmsApi.ts):** Reusable Axios API helper functions utilizing the pre-configured Axios instance with automatic JWT Bearer header injection.

2.3 Student Learning Experience

- **Student Dashboard (StudentDashboardPage.tsx):** Main portal with 4 real-time stat cards (Enrolled Subjects, Total Attempts, Highest Score, Average Percentage), quick quiz launcher, subject grid, and recent attempts log.
- **Subject Catalog & Chapter List (StudentSubjectCatalogPage.tsx & StudentChapterListPage.tsx):** Searchable academic subject catalog with chapter counters and structured chapter hierarchy views.
- **Material Viewer & PDF Stream (StudentMaterialViewerPage.tsx & PdfViewer.tsx):** Dual-tab viewer streaming binary PDF byte blobs from GET /api/documents/{id}/stream directly inside an embedded preview container.
- **Interactive Quiz Execution & Timer (StudentQuizExecutionPage.tsx & QuizTimer.tsx):** Single-question focus environment with Question Navigator grid, question state badges, countdown timer auto-submission, and submission modal confirmation.
- **Secure Answer Handling:** Quiz questions endpoint (GET /api/student/quizzes/{id}) delivers questions without revealing correct answers. Frontend maintains local Map of choices and submits only option indices.
- **Quiz Submission & Auto-Grading:** Submits choices to POST /api/student/quizzes/{id}/submit. Server evaluates answers against key and returns QuizResultDTO.
- **Quiz Result Page (StudentQuizResultPage.tsx):** Renders score, percentage, correct/wrong count, and attempt timestamp. Pass/Fail (isPassed) is rendered ONLY if returned by backend result DTO; no artificial threshold (e.g. 80%) is hardcoded.
- **Historical Performance Analytics (StudentProgressPage.tsx):** Interactive dashboard powered by chart.js featuring score percentage trend line chart, accuracy breakdown doughnut chart, and historical attempt records table.

2.4 Teacher Workspace

- **Material Upload Form (MaterialUploadForm.tsx):** Form supporting .pdf, .doc, .docx uploads up to 20MB with subject/chapter binding via POST /api/teacher/upload-pdf.
- **Manual Quiz Builder (ManualQuizBuilder.tsx):** Dynamic authoring form for title, time limit, and MCQ questions (Options A-D & correct answer selection) via POST /api/teacher/create-quiz (auto-published).
- **Teacher Dashboard (TeacherDashboardPage.tsx):** Tabbed workspace unifying Material Uploads, Manual Quiz Builder, and Content Overview table.

2.5 Routing & RBAC Integration

- **Routes & Navigation:** Registered routes in src/routes/index.tsx with ProtectedRoute guards enforcing ROLE_STUDENT and ROLE_TEACHER access control.

3. Technology Stack & Dependencies

Category	Technology / Library	Version	Purpose / Details
Backend Core	Java JDK	21	Execution runtime
Backend Framework	Spring Boot	3.2.5	Application framework & REST APIs
Database	PostgreSQL	16	Relational data store
Document Processing	Apache Tika & PDFBox	3.1.2	Text extraction from PDF/DOC/DOCX
Frontend Framework	React	19	SPA user interface framework
Language	TypeScript	5.x	Type safety & DTO interfaces
Build Tool	Vite	6.4.3	Frontend bundler & dev server
Charts & Viz	Chart.js & react-chartjs-2	4.4.8 / 5.3.0	Score trend line & accuracy doughnut charts
HTTP Client	Axios	1.8.2	REST API client with JWT interceptor

4. API Endpoints Reference

4.1 Reused Existing APIs

Method	Endpoint	Role	Description
GET	/api/student/progress	STUDENT	Fetches student stats & recent attempt records
GET	/api/subjects	ALL	Lists all active academic subjects
GET	/api/subjects/{id}	ALL	Retrieves subject details by ID
GET	/api/subjects/{id}/chapters	ALL	Retrieves chapters belonging to subject
GET	/api/documents/{id}/stream	ALL	Streams binary PDF content for preview
GET	/api/student/quizzes	STUDENT	Lists published student quizzes
GET	/api/student/quizzes/{id}	STUDENT	Fetches questions without revealing answers
POST	/api/student/quizzes/{id}/submit	STUDENT	Submits student choices for auto-grading
GET	/api/student/quizzes/results/{id}	STUDENT	Retrieves detailed graded result record
POST	/api/teacher/upload-pdf	TEACHER	Uploads materials (.pdf, .doc, .docx)
POST	/api/teacher/create-quiz	TEACHER	Authors & publishes manual MCQ quizzes

4.2 Genuinely New Backend APIs

Method	Endpoint	Role	Description
GET	/api/documents	ALL	Lists document metadata (filterable by subjectId & chapterId)
GET	/api/documents/{id}	ALL	Retrieves single document metadata by ID

5. Files & Components Summary

Files Added (13 total):

1. src/types/lms.ts – DTO TypeScript interfaces
2. src/api/lmsApi.ts – Centralized Axios API client functions
3. src/pages/student/StudentSubjectCatalogPage.tsx – Subject catalog page
4. src/pages/student/StudentChapterListPage.tsx – Chapter list page
5. src/pages/student/StudentMaterialViewerPage.tsx – Material & notes viewer page
6. src/components/PdfViewer.tsx – PDF streaming preview iframe component

- 7. src/components/QuizTimer.tsx – Countdown quiz timer component
- 8. src/pages/student/StudentQuizExecutionPage.tsx – Timed quiz execution environment
- 9. src/pages/student/StudentQuizResultPage.tsx – Graded quiz result view
- 10. src/pages/student/StudentProgressPage.tsx – Chart.js performance analytics page
- 11. src/components/teacher/MaterialUploadForm.tsx – Material upload form component
- 12. src/components/teacher/ManualQuizBuilder.tsx – Manual quiz authoring form component
- 13. src/test/java/com/learnpulse/backend/document/DocumentRetrievalIntegrationTest.java – Document list test

Files Modified (8 total):

- 1. DocumentService.java – Added getDocuments(subjectId, chapterId)
- 2. DocumentDownloadController.java – Added GET /api/documents & GET /api/documents/{id}
- 3. CourseProgressionIntegrationTest.java – Fixed tear-down table deletion order
- 4. BaseStatusControllerTest.java – Updated pgvectorInstalled matcher
- 5. StudentDashboardPage.tsx – Connected real backend endpoints & metrics
- 6. TeacherDashboardPage.tsx – Converted to tabbed workspace (Upload, Quiz Builder, Overview)
- 7. src/routes/index.tsx – Registered student & teacher routes with RBAC
- 8. MainLayout.tsx – Updated role-based header navigation links

6. Verification & Test Results

The complete backend Maven integration test suite runs **84 automated integration tests** with **0 failures**. Frontend TypeScript compilation (tsc -b) and Vite production bundling (vite build) completed with **0 errors**.

Test Class Name	Tests	Status	Scope / Area Tested
LearningAssistantApplicationTests	1	PASSED	Spring Context & Database Connectivity
JwtProviderTest	3	PASSED	JWT Generation, Signing, Validation
SecurityRbacIntegrationTest	8	PASSED	Authentication, CORS, RBAC Guards, Registration
DocumentIngestionIntegrationTest	9	PASSED	File Upload, Tika/PDFBox Extraction
DocumentRetrievalIntegrationTest	3	PASSED	Document Metadata & Filtering Endpoints
CourseProgressionIntegrationTest	10	PASSED	Chapter Locking, Progression & Unlocking Rules
AssessmentEngineIntegrationTest	39	PASSED	Quiz Submission, Auto-Grading & Scoring
AcademicHierarchyIntegrationTest	5	PASSED	Subject & Chapter CRUD Operations
BaseStatusControllerTest	1	PASSED	Health & System Status Controller
StudentTeacherProfileIntegrationTest	5	PASSED	Profile Updates & Role Metadata
TOTAL BACKEND SUITE	84	84 PASSED	BUILD SUCCESS (0 Failures, 0 Errors)

7. Scope Boundaries & Known Limitations

- **Pass/Fail Display Rule:** As per mentor specification, isPassed is displayed ONLY when explicitly supplied by backend result DTO. No artificial threshold (e.g. 80%) was added or hardcoded on the frontend.
- **Scope Exclusions:** AI question generation, RAG pipeline, and AI Tutor functionality are strictly out of scope for Week 8 and belong to Week 9+.
- **Git Operations:** No git commit or git push commands were performed as per instructions.
- **Architecture Preservation:** Preserved existing Week 7 Spring Security configuration, JWT authentication, and database schemas without redesign.

8. Sign-Off & Implementation Status

Verification Gate	Status	Details & Empirical Results
Backend Automated Tests	PASSED	84/84 tests passed (mvn test)
Frontend Production Build	PASSED	Built cleanly with 0 TypeScript errors (npm run build)
End-to-End Workflow	VERIFIED	Teacher Upload & Quiz Build -> Student Browse, PDF Stream, Timed Execution, Grading, & Analytics
Mentor Requirements	SATISFIED	Strictly aligned with Week 8 specification guidelines